Yoonseon Choi

Contact Information

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Fusion Tech Center (FTC), Hanyang University, Seoul, Republic of Korea, 04763

Research Interest

My research focuses on multimodal AI models that precisely infer human states and contextual information from time-series sensor data. I investigate methods for analyzing and refining real-world data collected from users to develop domain-specific personalized prediction models. Based on these research capabilities, I have experience building AI systems for real industrial applications.

Human-Computer Interaction | Times Series Forecasting | Digital Healthcare

Education

Hanyang University, Seoul, Republic of Korea — *Feb.2026*
M.S. in Department of Data Science
GPA: 4.25 / 4.50

Hanyang University, Seoul, Republic of Korea — *Feb. 2024*
B.S. in Department of Data Science
GPA: 4.28 / 4.50 (*Summa Cum Laude*)

Work Experience

Researcher, *Hanyang University Industry-University Cooperation Foundation* — *Present*

AI Engineer Intern, *CJ Logistics TES Lab* — *Aug. 2025 - Nov. 2025*

· Developed time-series and tabular models for logistics demand forecasting

Publications

International Conference and Journal Papers (* indicates equal contributions)

[4] *(Under Review)* VRS-LLM: Predicting Virtual Reality Sickness Using Large Language Models Based on Multimodal Time Series Data
Y Choi, D Jeong, B Kim, and K Han
IJHCI (*International Journal of Human-Computer Interaction*) (SCIE Journal, Q1; IF=4.9)
Links: Paper

[3] VR-ACT: Transforming Clinical Evidence of Acceptance and Commitment Therapy through Virtual Reality for Subthreshold Depression
B Kim, D Jeong, **Y Choi**, H Kim, and K Han
Springer Virtual Reality (SCIE Journal 2025, Q1; IF=5.0)
Links: DOI | Paper

[2] Enhancing Mindfulness-Based Cognitive Therapy in Virtual Reality: A Prospective Interventional Study
{B Kim*, D Jeong*}, **YS Choi**, YY Choi, H Kim, and K Han
Scientific Reports (SCIE Journal 2025, Q1; IF=3.8)
Links: DOI | Paper

[1] Poster: Early Prediction of Cybersickness in Virtual Reality Using a Large Language Model for Multimodal Time Series Data
Y Choi, D Jeong, B Kim, and K Han
UbiComp 2024 (*The ACM International Joint Conference on Pervasive and Ubiquitous Computing*)
Links: DOI | Paper | Video

Domestic Conference and Journal Papers (In Korean, Lightly Peer Reviewed)

[5] Development of a Personalized Early Prediction Model for Cybersickness in Virtual Reality
Y Choi, D Jeong, B Kim, and K Han
KTCP (*KIISE Transactions on Computing Practices*, KCI Journal 2025)
Invited paper from KCC 2024
Links: DOI | Paper

[4] Virtual Reality-Based Mindfulness for Depression Alleviation: Focusing on Content Development and Data Analysis
Y Choi, B Kim, D Jeong, H Kim, and K Han
HCI KOREA 2025
Links: DOI | Paper

[3] Poster: Development of a Personalized Early Prediction Model for Cybersickness in Virtual Reality Using a Large Language Model
Y Choi, D Jeong, B Kim, and K Han
KDMS 2024 *(Korea Data Mining Society, Autumn Conference)*
Grand Paper Award
Links: Poster | Award

[2] Poster: A Study on the Effectiveness of VR-Based Mindfulness-Based Cognitive Therapy Using Multimodal Data
D Jeong, B Kim, **Y Choi**, and K Han
KDMS 2024 *(Korea Data Mining Society, Autumn Conference)*
Best Paper Award
Links: Poster | Award

[1] Early Prediction of Cybersickness in Virtual Reality Using a Long-term Time Series Forecasting Model
Y Choi, D Jeong, B Kim, and K Han
KCC 2024 *(Korea Computer Congress)*
Best Presentation Award
Links: DOI | Paper | Award

Awards & Honors

Grand Paper Award, *Korea Data Mining Society*	*Fall 2024*
Best Paper Award, *Korea Data Mining Society*	*Fall 2024*
Best Presentation Award, *Korea Computer Congress*	*2024*
Summa Cum Laude, *Hanyang University*	*2024*
Academic Excellence Award, *Hanyang University*	*2023*
Academic Excellence Scholarship, *Gwangju Citizen Scholarship*	*2020-2021*
Academic Excellence Award, *Dongduk Women's University*	*2019-2020*

Patent

Filed Patent

- Method and system for predicting changes in logistics demand — *Sep. 2025*
  Application number: 10-2025-0140350
- An LLM-based AI model that reflects inter-modality relationships in multimodal time series data for personalized prediction of VR sickness — *Jul. 2025*
  Application number: 10-2025-0100532

ACADEMIC SERVICE

Conference Reviewer

- The IEEE International Conference on Data Science and Advanced Analytics (**DSAA**) *2025*
- Korea Computer Congress (**KCC**) *2025*
- The Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies (**IMWUT**) *2025*
- The HCI Society of Korea (**HCIK**) *2025*
- The Conference on Information and Knowledge Management (**CIKM**) *2024*

Journal Reviewer

- Scientific Reports *2024*

TEACHING EXPERIENCE

Lecture Assistant

- Data Analytics Expert Training Program, ***Hanwha General Insurance*** *Jun. 2025*

Teaching Assistant (TA)

- Data Science Fundamentals *Spring 2025*
- Data Science Fundamentals *Spring 2024*

CERTIFICATION

ADsP (Advanced Data Analytics Semi-Professional), *Korea Data Agency* *Mar. 2023*

REFERENCES

Kyungsik Han, *Associate Professor* kyungsikhan@hanyang.ac.kr
Hanyang Human-Centered Computing Lab

Itaeck Joo, *Collaboration Professor* itjoo@hanyang.ac.kr
Industrial AI Talent Development Bootcamp Program

Bogoan Kim, *Assistant Professor* bogoan@cbnu.ac.kr
CBNU HCI Lab